

Chapter 8

Finite State Machine Optimization

8.1 Objectives

1. Analyse and design both Mealy and Moore sequential circuits with multiple inputs and multiple outputs
2. Convert between Mealy and Moore designs
3. Perform a state assignment using the guideline method
4. Reduce the number of states in a state table using row reduction and implication tables

8.2 Mealy vs Moore Finite State Machines

Definition 8.1 (Finite State Machines (FSM)). [1, Sec 3.4] A FSM has M inputs, N outputs and k bits of unique states (2^k states).

Definition 8.2 (Mealy FSM). [1, Sec 3.4.3] Named after George M. Mealy, in Mealy FSM the output depends both on inputs and the states.

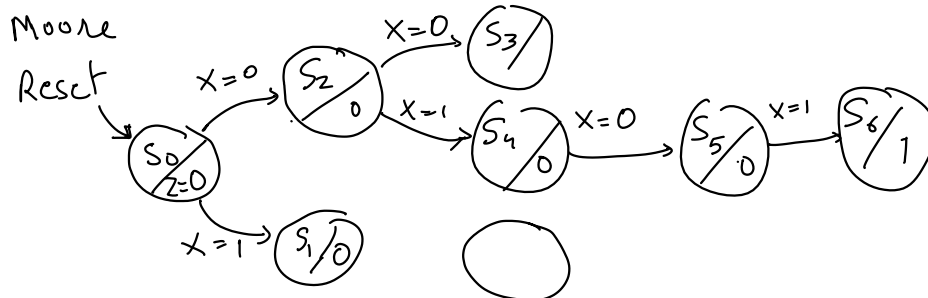
Definition 8.3 (Moore FSM). [1, Sec 3.4.3] Named after Edward F. Moore, in Moore FSM the output depends only on the states.

Example 8.1. A sequential circuit has one input (X) and one output (Z). The circuit examines groups of four consecutive inputs and produces an output $Z=1$ if the input sequence 0010 or 0001 occurs. The sequences can overlap. Draw both Mealy and Moore timing diagrams. Find the Mealy and Moore state graph.

Example 8.2. A sequential circuit has one input (X) and one output (Z). The circuit examines groups of four consecutive inputs and produces an output $Z=1$ if the input sequence 0101 or 1001 occurs. The circuit resets after every four inputs. Draw both Mealy and Moore timing diagrams. Find the Mealy and Moore state graph.

Example seq

$X = \underline{0010} \underline{1001} \underline{0100} \underline{0101} \underline{1001}$
 $Z = 0000 \ 0000 \ 0000 \ 0001 \ 0001$



What are we using to represent the state?

Define

$S_0 = \begin{matrix} * & * & * & * \end{matrix}$
 $S_2 = \begin{matrix} * & * & * & 0 \end{matrix}$
 $S_1 = \begin{matrix} * & * & * & 1 \end{matrix}$
 $S_3 = \begin{matrix} * & 1 & 0 & 0 \end{matrix}$
 $S_4 = \begin{matrix} * & 1 & 0 & 1 \end{matrix}$
 $S_5 = \begin{matrix} * & 0 & 1 & 0 \end{matrix}$
 $S_6 = \begin{matrix} 0 & 1 & 0 & 1 \end{matrix}$

8.3 Full procedure for designing sequential logic circuit

1. Convert the word problem to a state transition diagram. Let the states be $S_0, S_1, S_2, \dots, S_n$.

2. Draw state transition table with named states. For example,

Present State	Next State		Outputs	
	$X=0$	$X=1$	$X=0$	$X=1$
S_0	S_1	S_2	0	0
S_1	S_2	S_0	0	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

3. State reduction step: Reduce the number of required states to a minimum. Eliminate unnecessary or duplicate states.

4. State assignment step: Assign each state a binary representation. For example,

State name	State assignments ($Q_2Q_1Q_0$)
S_0	000
S_1	001
$\vdots$	$\vdots$

5. Draw State assigned transition table. For example,

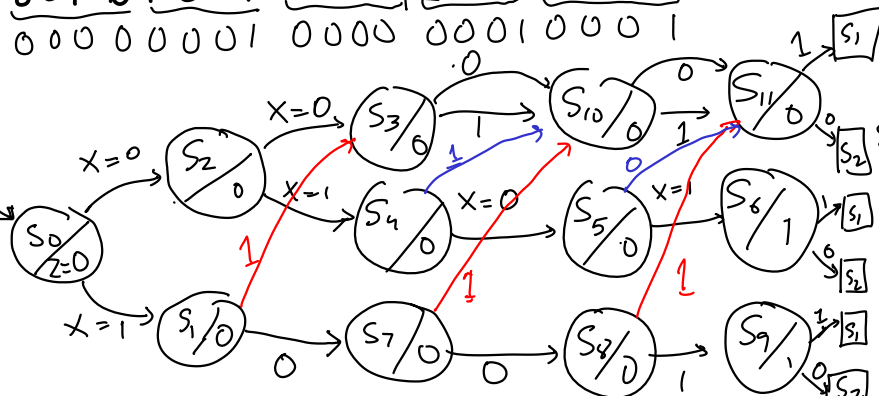
Inputs (X_1X_0)	Present State (Q_1Q_0)	Next State ($Q_1^+Q_0^+$)	Outputs (Z_1Z_0)
0 0	00	01	0 0
0 0	01	10	0 0
$\vdots$ $\vdots$	$\vdots$	$\vdots$	$\vdots$ $\vdots$

Example seq

$X = 0010100101001001$
 $Z = 000000010000000100001$

0101
1001

Moore
Reset



Define

$S_0 = ****$
 $S_2 = ***0$
 $S_1 = ***1$
 $S_3 = **00$
 $S_4 = *001$
 $S_5 = *010$
 $S_6 = *101$
 $S_7 = *100$
 $S_9 = 1001$

12-states

4-bit register
4 ff

$S_{11} = 4 \text{ bits}$
not(0101/1001)
 $S_2 = 2 \text{ bits}$
not(01/10)

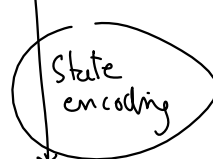
$S_{10} = 3 \text{ bits}$
not(010/100)

What are we using to represent the state?

Prob statement



FSM



Circuit

State encoding

$Q_3 Q_2 Q_1 Q_0$
 $S_0 = 0000$
 $S_1 = 0001$

State encoded trans. table

Inputs	State			
X	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
0	0	0	0	1

Next State

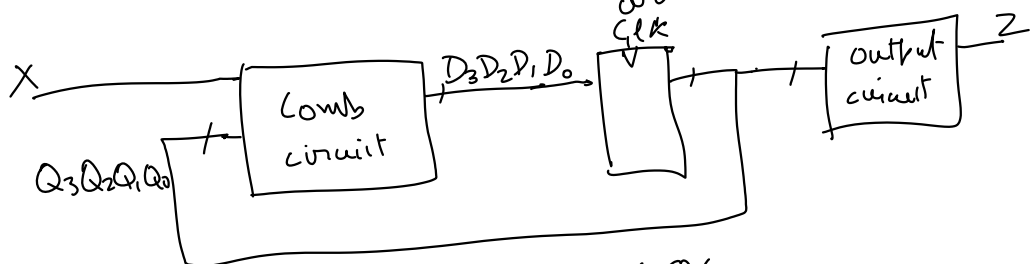
$Q_3^+ Q_2^+ Q_1^+ Q_0^+$

$D_3 D_2 D_1 D_0$ Truth table

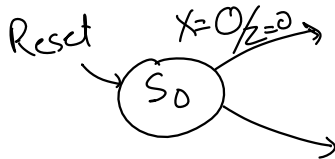
Output

Z

Don't cares
are allowed



Mealy state graph



- (a) Use excitation tables to find truth tables for the combinational circuits. For example, the excitation table for J-K ff is

Q	Q ⁺	J	K
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

8.4 State assignment by guideline method [2, Section 8.2.5]

8.4.1 State Maps

Example 8.3. Draw a state map for a sequential assignment of the states

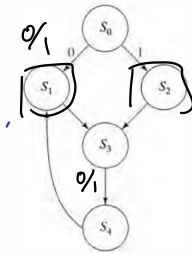
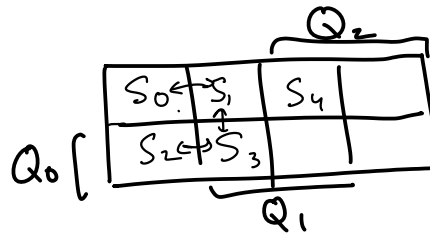


Figure 8.27 Five-state finite state machine.



$Q_2 \quad Q_1 \quad Q_0$
 $S_0 \quad ? \quad ? \quad ?$

State map: a K-map with states in it
 Guideline method: guides which states should be next to each other.

8.4.2 Guideline method

Guideline method states that the following states should be adjacent in the state map according to the following priorities:

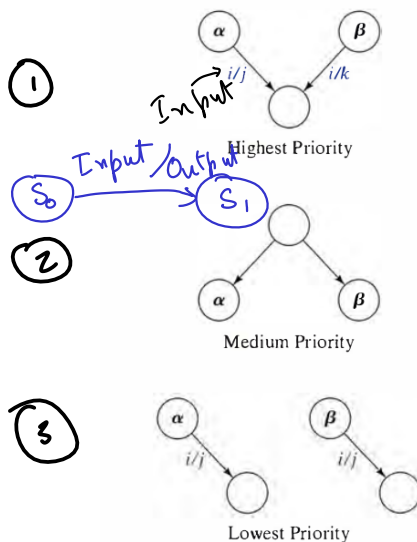


Figure 8.29 Adjacent assignment priorities.

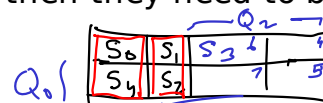
Put states next to each other if they lead to the same next state with same inputs

(S_0, S_1) (S_1, S_2)

if the states are coming out of the same state, then put them next to each other with medium priority

(S_1, S_2)

if the states have the same output to the same inputs then they need to be next to each other.



Example 8.4. A state transition table is given. Find optimal state assignment by using the guideline method.

	Q_2	Q_1	Q_0
S_0	0	0	0
S_4	0	0	1
S_1	0	1	1
S_2	0	1	0
S_3	1	1	0

1. Highest priority: Same next state for same inputs
2. Medium priority: Same prev state
3. Lowest priority: Same output for the same inputs

① (S_3, S'_4)

② (S'_3, S'_4)

③ $(S_0, S'_1, S'_3) (S_0, S'_1, S'_3, S'_4)$

$$Q_1 \begin{array}{|c|c|} \hline S_0^0 & S'_3^2 \\ \hline S'_1^{11} & S'_4^3 \\ \hline \end{array}$$

Input Sequence	Present State	Next State		Output	
		X=0	X=1	X=0	X=1
Reset	S_0	S'_1	S'_1	0	0
0 or 1	S'_1	S'_3	S'_4	0	0
00 or 10	S'_3	S_0	S_0	0	0
01 or 11	S'_4	S_0	S_0	1	0

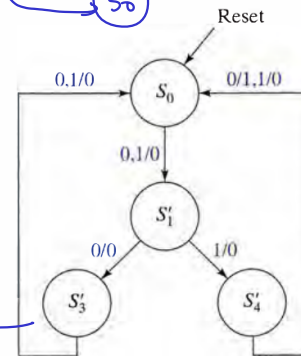
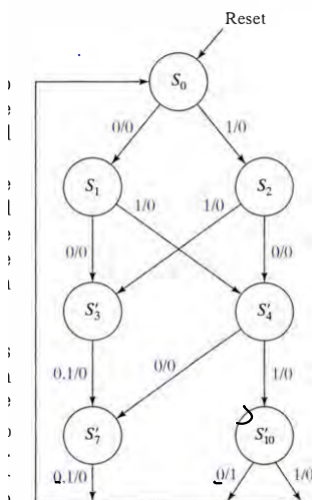
$$\begin{array}{c|cc} Q_1 & Q_0 & \\ \hline S_0 & 0 & 0 \\ S'_1 & 0 & 1 \\ S'_3 & 1 & 0 \\ S'_4 & 1 & 1 \end{array}$$


Figure 8.30 Reduced state diagram for 3-bit sequence detector.

Example 8.5. Draw a Mealy FSM for detecting binary string 0110 or 1010. The machine returns to the reset state after each and every 4-bit sequence. Draw the state transition diagram on your own as practice problem. The state transition diagram is given here. Find optimal state assignment by using the guideline method.



① HP: $(S_3, S'_4), (S_7, S'_{10})$

② MP: $(S_1, S_2), (S_3, S'_4), (S_7, S'_{10})$

③ LP: $(S_0, S_1, S_2, S_3, S'_4, S_7, S'_{10})$

State map → State assignment

$$Q_2 \begin{array}{|c|c|c|c|} \hline S_3^0 & S'_1^2 & S_7^6 & S'_4^7 \\ \hline S'_4^1 & S'_2^3 & S'_{10}^5 & \\ \hline \end{array}$$

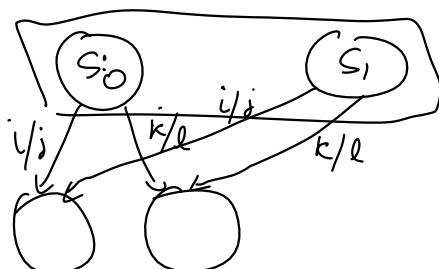
$$\begin{array}{c|ccc} & Q_2 & Q_1 & Q_0 \\ \hline S_0 & 1 & 0 & 0 \\ S_1 & 0 & 1 & 0 \\ S_2 & 0 & 1 & 1 \\ S_3 & 0 & 0 & 0 \\ S_4 & 1 & 0 & 0 \\ S'_1 & 1 & 1 & 1 \\ S'_{10} & 1 & 1 & 1 \end{array}$$

8.5 State reduction by implication chart

Example 8.6. Design a Mealy FSM for detecting binary sequence 010 or 0110. The machine returns to reset state after each and every 3-bit sequence. For now the state transition table is given. Reduce the following state transition table

When is a state duplicate or redundant?

Answer: If it leads to the same next state and output under for each possible input



What is a state? States in particular markovian states are memory states. A state captures all the past information that is needed for future decision making

State definition

Input Sequence	Present State	Next State		Output	
		X=0	X=1	X=0	X=1
Reset	S_0	S_1	S_2 S_1	0	0
0	S_1	S_3	S_4	0	0
1	S_2 S_1	S_5 S_3	S_6 S_4	0	0
00	S_3	S_0	S_0	0	0
01	S_4	S_0	S_0	1	0
10	S_5 S_3	S_0	S_0	0	0
11	S_6 S_4	S_0	S_0	1	0

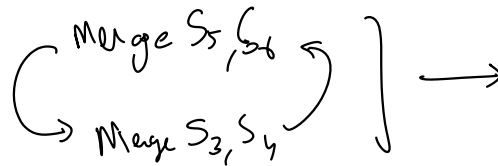
Can S_4, S_6 be merged? Yes

$$S_4 = (01/11) = *1$$

S_3, S_5

Yes. Replace S_5 with S_3

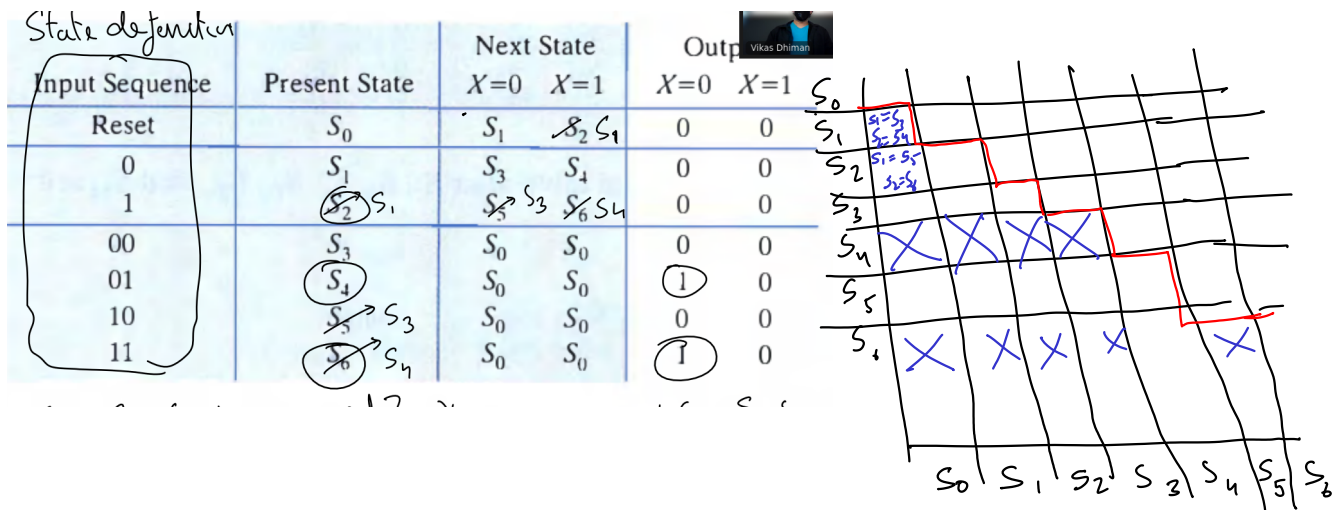
Can S_1, S_2 be merged?
Yes,



8.5.1 Implication chart Summary

The algorithms for state reduction using the implication chart method consists of the following steps

1. Construct the implication chart, consisting of one square for each possible combination of states taken two at a time.
2. For each square labeled by states S_i and S_j , if the outputs of the states differ, mark the square with an X; the states are not equivalent. Otherwise, they may be equivalent. Within the square write implied pairs of equivalent next states for all input combinations.
3. Systematically advance through the squares of the implication chart. If the square labeled by states S_i, S_j contains an implied pair S_m, S_n and square S_m, S_n is marked with an X, then mark S_i, S_j with an X. Since S_m, S_n are not equivalent, neither are S_i, S_j .
4. Continue executing Step 3 until no new squares are marked with an X.
5. For each remaining unmarked square S_i, S_j , we can conclude that S_i, S_j are equivalent.



Review

8.6 Study guide

8.6.1 Midterm 1

- ✓ Binary numbers, Hexadecimal, Sign-magnitude, One's-complement and Two's complement. Conversions between them.
- ✓ Generate minterms, maxterms, SOP canonical form and POS canonical forms and convert between them
- ✓ Understand and use the laws and theorems of Boolean Algebra
- ✓ Perform algebraic simplification using Boolean algebra
- ✓ Simplification using K-maps (upto 5-input variables)
- ✓ Derive sum of product and product of sums expressions for a combinational circuit
- ✓ Convert combinational logic to NAND-NAND and NOR-NOR forms

8.6.2 Midterm 2

- ✓ Design Hazard-free two level circuits.
 - ✓ Different between and the limitations of level-triggered latches and edge-triggered flip-flops.
 - ✓ Understand the difference between synchronous and asynchronous inputs
 - ✓ Derive a state graph or state table from a word description of the problem
 - ✓ Implement a design using JK, SR, D or T flip-flops
 - ✓ Analyze a sequential circuit and derive a state-table and a state-graph
- Convert between Mealy and Moore designs
- Partition a system into multiple state machines

8.6.3 Final (includes previous topics)

- ☒ Reduce the number of states in a state table using row reduction and implication tables
- ☒ Perform a state assignment using the guideline method
- ☒ Analyse and design both Mealy and Moore sequential circuits with multiple inputs and multiple outputs
- ☒ Design combinational circuits using multiplexers and decoders
- ☒ Know fan-out and noise margin
- ☒ Know the differences and similarities between PAL, PLA, and ROMs (☒ and can use each for logic design)
- ☒ Simplification using Quine-McCluskey method
- ☒ Design combinational circuits for positive and negative logic
- ☒ Design a CMOS complex gate
- ☒ Compute noise margin of one device
- ☒ Describe how tri-state and open-collector outputs are different from totem-pole outputs.

Chapter 9

Sample Midterm 2

Student Name:

Student Email:

9.1 Instructions

- There are five problems. All problems are required.
- Maximum number of marks is 50. This exam amounts 10% toward the final grade.
- Time allowed is 50 minutes.
- In order to minimize distraction to your fellow students, you may not leave during the last 10 minutes of the examination.
- The examination is closed-book. One 8×11 in two-sided cheatsheet is allowed.
- Non-programmable calculators are permitted.
- Please use a pen or heavy pencil to ensure legibility. Colored pens/pencils are recommended for K-map grouping.
- Please show your work; where appropriate, marks will be awarded for proper and well-reasoned explanations.

Problem 9.1. Table 9.1 shows the truth table for a BCD to 7-segment display. The inputs corresponding to the missing rows in the truth table should be considered as don't care. Implement segment "a" using a 4:1 MUX (multiplexer) and one other gate. (10 marks)

Row	w_3	w_2	w_1	w_0	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1

Table 9.1: Truth table for BCD to seven-segment display. The missing combinations of inputs should be considered as don't care.

Problem 9.2. Draw Mealy state transition diagram which investigates an input sequence X and will produce an output of $Z = 1$ for any input sequence ending in 0011 or 110.

Example:

$X =$	1	1	0	0	1	0	0	1	0	1	0	0	1	0	1
$Z =$	0	0	0	1	0	1	1	0	1	0	0	1	0	1	0

Notice that the circuit does not reset to the start state when an output of $Z = 1$ occurs. A minimum solution requires six states. Assign 000 to the start state. (20 marks).

Problem 9.3. Find the expression for J_0 and K_0 assuming that J_0 and K_0 are inputs to the J-K flip flop that capture the state of the second most significant bit Q_0 of the following state encoded table. The state encoding table given with state encoding denoted as $Q_2Q_1Q_0$. (20 marks).

Present State $Q_2Q_1Q_0$	Next State		Output	
	$X = 0$ $Q_2^+Q_1^+Q_0^+$	$X = 1$ $Q_2^+Q_1^+Q_0^+$	$X = 0$ Z	$X = 1$ Z
000	100	101	1	0
001	100	101	0	1
010	000	000	1	0
011	000	000	0	1
100	111	110	1	0
101	110	110	0	1
110	011	010	1	0
111	011	011	0	1

Problem 4. A sequential circuit is to be used to control the operation of a vending machine that dispenses a \$0.25 product. The circuit has three inputs (N, D, and Q) and two outputs (R and C). The coin detector mechanism in the vending machine is synchronized with the same clock as the sequential circuit you are to design. The coin detector outputs a single 1 to the N, D, or Q input for every nickel (5 cents), dime (10 cents), or quarter (25 cents), respectively, that the customer inserts. Only one input will be 1 at a time. When the customer has inserted at least \$0.25 in any combination of nickels, dimes, and quarters, the vending machine must give change and dispense the product. The coin return mechanism gives change by returning nickels to the customer. For every 1 output on C, the coin return mechanism will return one nickel to the customer. The product is dispensed when the circuit outputs a single 1 on output R. The circuit should reset after dispensing the product.

Example: The customer inserts a nickel, a dime, and a quarter. The circuit inputs and outputs could look like this:

Inputs:

N= 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0
 D= 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0
 Q= 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0
 R= 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0
 C= 0 0 0 0 0 0 0 0 0 0 0 1 1 1 0 0 0

Note that any number of 0's can occur between 1 inputs. Complete the following Moore state table for the sequential circuit, and for each state indicate how much money the customer has inserted or how much change is due. Use the following header for the state table (PS means Present state)

Money inserted	Change due	PS	Next State				Output (RC)
		NDQ = 000 NDQ = 100 NDQ = 010 NDQ = 001					
0	0	S ₀	S ₀	S ₁	S ₂	S ₃	00

(30 marks)